Programming Worlds

PROGRAMMING WORLDS ACADEMY

Git & GitHub 101

From Zero to Your First Pull Request

BEYOND CODE. BEYOND LIMITS.

03

MODULE 03

Creating Your First Repository

Turn a plain folder into a Git repository and watch Git notice your first files.

MODULE 03 · LESSON 3.1

Understanding repositories

فهم المستودعات

Learning objectives

- Define a repository.
- Tell a project folder from a Git repository.
- Explain what the `.git` folder holds.
- Tell local from remote repositories.

Prerequisites

Module 2 completed.

Introduction

You have heard the word “repository” many times. Here it gets a precise meaning, and you will see exactly what makes a folder special to Git.

Explanation

A **repository** (repo) is a project folder that Git is tracking, together with the full history of that project.

A **normal folder** is just files. It becomes a **Git repository** when you run `git init` inside it. Git then creates a hidden folder named `.git`.

The **.git directory** stores the commits, branches and settings of the repository. Your own files live next to it, not inside it. **Never edit or delete .git by hand.** Deleting it removes the history, although your files stay.

A **local repository** is the one on your computer. A **remote repository** is a copy on a server such as GitHub. You will connect them in Module 5.

To see hidden items in Windows, open File Explorer and choose **View → Show → Hidden items**.

Real-world analogy

> A folder is a house. The `.git` folder is the archive of security-camera recordings: the house looks the same, but the archive remembers everything that happened inside.

Diagram

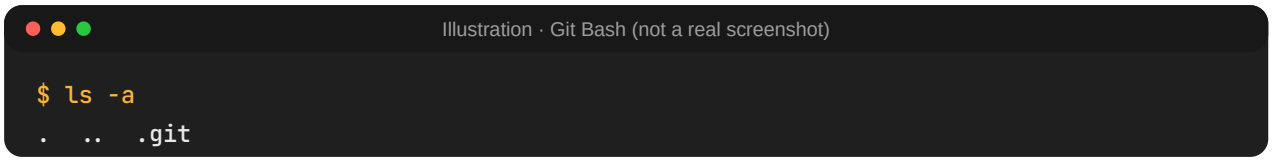


Figure 1 · Illustration: a repository contains a hidden .git folder

Commands

In Git Bash, `ls -a` lists all files, including hidden ones. Inside a repository you will see `.git`.

```
ls -a
```

Common mistakes and solutions

Deleting the .git folder.	Your history is gone. Only do this if you truly want to stop using Git in that project.
Thinking .git is inside your project files.	It is a hidden sibling of your files. Your files stay outside it.
Creating a repo inside another repo.	Check you are in the right folder before `git init`.

Practical exercise

> Turn on hidden items in File Explorer. Open your Documents folder and check that it does not contain a `.git` folder.

Expected result

You see no `.git` folder. Ordinary folders are not repositories until you initialize them (Lesson 3.2).

Knowledge check

1. What makes a folder a Git repository?

- ○ Its name
- ○ A hidden .git folder inside it
- ○ A README file

2. What does .git store?

- ○ Commits, branches and settings
- ○ Your passwords
- ○ Windows updates

3. Where is a remote repository?

- ○ In the Recycle Bin
- ○ On a server such as GitHub
- ○ Inside .git

Answers

- A hidden .git folder inside it — git init creates .git, which holds the history.
- Commits, branches and settings — It holds the repository data.
- On a server such as GitHub — Remote means a copy hosted elsewhere.

Summary

A repository is a project folder plus its Git history, stored in the hidden .git folder. Local is on your computer, remote is on a server such as GitHub.

Key vocabulary

Repository	A project tracked by Git with its history
.git	Hidden folder holding the history and settings
Local / Remote	On your computer / on a server

Next lesson

Lesson 3.2 — git init and git status. Next you create a repository yourself.

MODULE 03 · LESSON 3.2

git init and git status

git init ٩git status

Learning objectives

- Create a folder and enter it from the terminal.
- Initialize a repository with `git init`.
- Read the output of `git status`.

Prerequisites

Lesson 3.1 and a configured Git (Lesson 2.3).

Introduction

Two commands begin every Git project: `git init` creates the repository, and `git status` tells you what Git sees. You will run `git status` hundreds of times.

Explanation

`mkdir my-first-project` makes a new folder. `cd my-first-project` moves into it. Choose a location such as Documents. Do not use a folder that already holds other projects.

`git init` turns the current folder into a repository. Git answers “Initialized empty Git repository in …/.git/”. Running it again is harmless: Git says it reinitialized.

`git status` shows the branch, whether there are commits, and which files are untracked, modified or staged.

In the output, **On branch main** is the current branch. **No commits yet** means the history is empty. **nothing to commit** means Git sees no files yet.

Real-world analogy

> `git init` is switching on the recorder. `git status` is asking the recorder what it currently sees.

Diagram

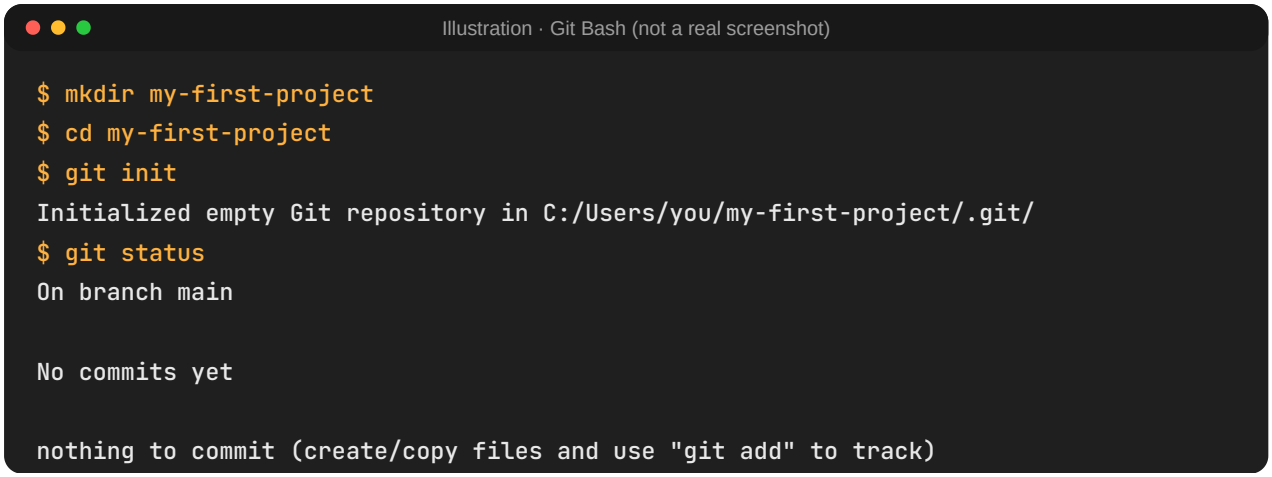


Figure 1 · Illustration of the expected output

Commands

`mkdir my-first-project`
`cd my-first-project`
`git init`
`git status`

Expected output: see Figure 1. The path in your output will be your own.

Common mistakes and solutions

"fatal: not a git repository"	You are outside a repository. Use `cd` to enter the project folder, or run `git init` first.
Running git init in the wrong folder.	Run `pwd` to see where you are before you initialize.
Typing the folder name with spaces.	Use names without spaces, such as `my-first-project`.

Practical exercise

> Create `my-first-project`, initialize it, and run `git status`.

Expected result

Git reports an empty repository on branch `main` with no commits yet.

Knowledge check

1. Which command creates a repository?

- ○ git new
- ○ git init
- ○ git start

2. What does “No commits yet” mean?

- ○ The history is empty
- ○ Git is broken
- ○ The repo is private

3. Which command shows what Git sees?

- ○ git look
- ○ git status
- ○ git show-all

Answers

- git init — git init initializes a repository.
- The history is empty — No commit has been made.
- git status — git status reports the working state.

Summary

You created a folder, ran git init to make it a repository, and read the result of git status.

Key vocabulary

git init	Create a repository in the current folder
git status	Show the state of files and branch
pwd	Print the current folder

Next lesson

Lesson 3.3 — Tracking your first files. Now you add real files and watch Git notice them.

MODULE 03 · LESSON 3.3

Tracking your first files

تتبّع ملفاتك الأولى

Learning objectives

- Create the three starter files.
- See untracked files in `git status`.
- Name the four file states.

Prerequisites

Lesson 3.2.

Introduction

Git sees your folder but does not track anything until you tell it to. Here you create the first files and watch how Git reports them.

Explanation

In VS Code choose **File → Open Folder** and open `my-first-project`. Create three files: `index.html`, `style.css` and `README.md`. Paste the contents below and **save each file**. A dot on a tab means it is not saved yet.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>My First Project</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Hello, Git!</h1>
  <p>This is my first project.</p>
</body>
</html>
```

```
body {
  font-family: sans-serif;
  margin: 2rem;
}
```

```
h1 {
  color: #3B41E3;
}
```

```
# My First Project

A small website I use to learn Git and GitHub.
```

Run `git status`. Git lists the three files under **Untracked files**: it sees them but is not tracking them.

The four states: Untracked (new, unknown to Git), **Modified** (tracked file changed), **Staged** (chosen for the next commit), **Committed** (safely stored in history). In VS Code, untracked files show a **U** beside the name and modified files show an **M**. You will stage and commit them in Module 4.

Real-world analogy

> Untracked files are guests who walked into the house. Git sees them, but nobody has put them on the guest list yet.

Diagram

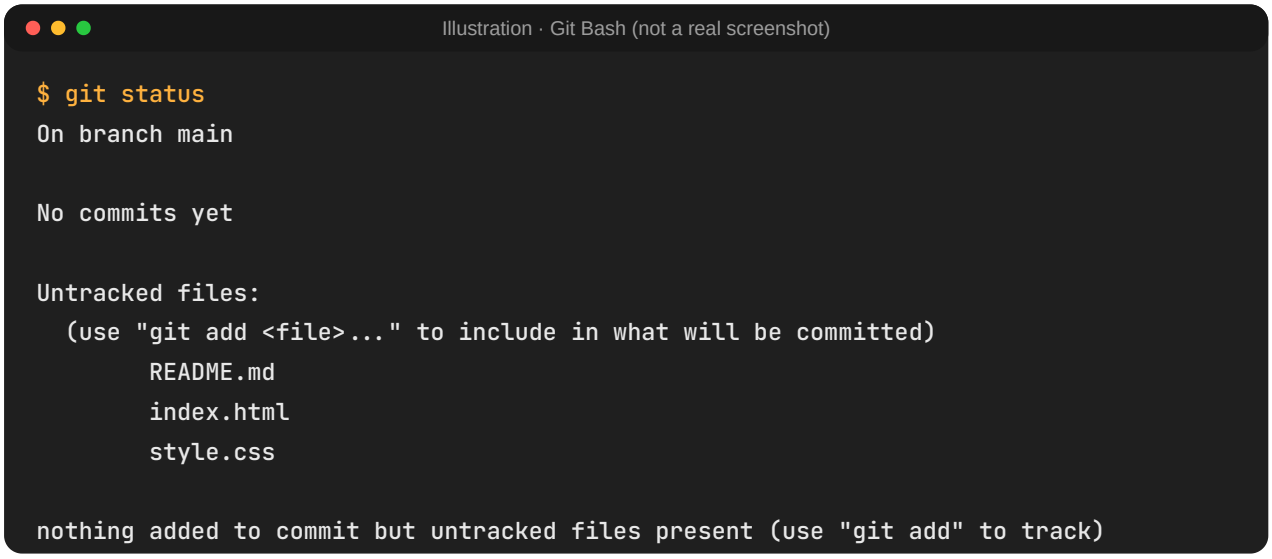


Figure 1 · Illustration of the expected output

Commands

`git status`

Run it after creating and saving the files. You can also copy the finished files from the course folder `starter-project/` .

Common mistakes and solutions

Files not saved.	Save with `Ctrl+S` . Git only sees saved changes.
Files created outside the project folder.	Check they are inside `my-first-project` .
Expecting git status to track the files.	It only reports. Tracking starts with git add.

Practical exercise

> Create the three files, save them, and run `git status` .

Expected result

All three files appear under “Untracked files”, as in Figure 1.

Knowledge check

1. What state is a brand-new file?

- ○ Pushed
- ○ Untracked
- ○ Committed

2. What does U mean in VS Code?

- ○ Untracked file
- ○ Updated online
- ○ Undo

3. Which state comes after staged?

- ○ Deleted
- ○ Committed
- ○ Untracked

Answers

- Untracked — Git has not seen it before.
- Untracked file — U marks an untracked file.
- Committed — Commit stores staged changes in history.

Summary

You created index.html, style.css and README.md, saw them as untracked, and learned the four file states.

Key vocabulary

Untracked	Not yet tracked by Git
Modified	A tracked file changed since the last commit
Staged	Marked for the next commit
Committed	Stored in the history

Next lesson

> **Module 4 — Git add, commit and history.** The most important module: you will save your first snapshots.

MODULE 03

Practical lab 3 · Create an HTML project and initialize Git

1. Create the folder `my-first-project` and enter it.
2. Run `git init`.
3. Open the folder in VS Code and create `index.html`, `style.css` and `README.md` (copy from `starter-project/`).
4. Save all files and run `git status`.

Check	Pass if…
Repository	`ls -a` shows `.git`
Files	Three files exist and are saved
Status	All three are listed as untracked

Evaluation: 3 of 3 checks = pass. Keep this folder: you will use it in every later module.

Module 3 quiz (pass mark: 70%)

1. Which command creates a repository?

- ○ git add
- ○ git push
- ○ git init

2. What does .git contain?

- ○ Your password
- ○ The repository history and settings
- ○ Your HTML files

3. What does git status show first?

- ○ The current branch
- ○ Your email
- ○ The weather

4. A new file shows as…

- ○ Merged
- ○ Untracked
- ○ Committed

5. Why save files before git status?

- ○ Windows requires it
- ○ To upload them
- ○ Git only sees saved changes

Answers

- git init — git init initializes a repository.
- The repository history and settings — It is the repository database.
- The current branch — The first line names the branch.
- Untracked — Git has not tracked it yet.
- Git only sees saved changes — Unsaved edits are not on disk.